

MIP Formulations for Delete-Free AI Planning

AIROYoung Workshop, 2026

AI Planning : Background & Notation

$$\Pi = \langle P, A, I, G, \text{cost} \rangle$$

- P : Set of boolean variables (facts)
- A : Set of actions, each with
 - $\text{pre}(a) \subseteq P$: Preconditions
 - $\text{add}(a) \subseteq P$: Add effects
 - $\text{del}(a) \subseteq P$: Delete effects
- $I, G \subseteq P$: Initial and Goal state
- $\text{cost} : A \rightarrow \mathbb{R}_0^+$: Non-negative cost function

AI Planning : Delete-Free relaxation

- Usually solved with the A^* algorithm
- A^* needs an admissible heuristic
- Delete-Free Relaxation (h^+): drop the delete effects $\rightarrow$ makes for an admissible heuristic
- $h^{\max}, h^{\text{add}}, h^{\text{LM}}$ are other admissible heuristics $\rightarrow$ all lower bounds of h^+
- Optimal plans for general planning tasks can be computed by iteratively solving and reformulating delete-relaxed tasks
- There exists combinatorial problems which can be modeled as delete-free problems
- h^+ is only NP-hard: we can use Mixed Integer Programming

MIP Formulation : Basic model

$$\min \sum_{a \in A} \text{cost}(a) x_a$$

$$\sum_{a \in A: p \in \text{add}(a)} x_{a,p} = x_p$$

$$\sum_{a \in A: p \in \text{pre}(a), q \in \text{add}(a)} x_{a,q} \leq x_p$$

$$x_{a,p} \leq x_a$$

$$x_p = 1$$

$$\forall p \in P$$

For a fact to be used, it needs to be achieved by an action

$$\forall p, q \in P$$

For an action to be used, its preconditions need to be achieved

$$\forall a \in A, p \in \text{add}(a)$$

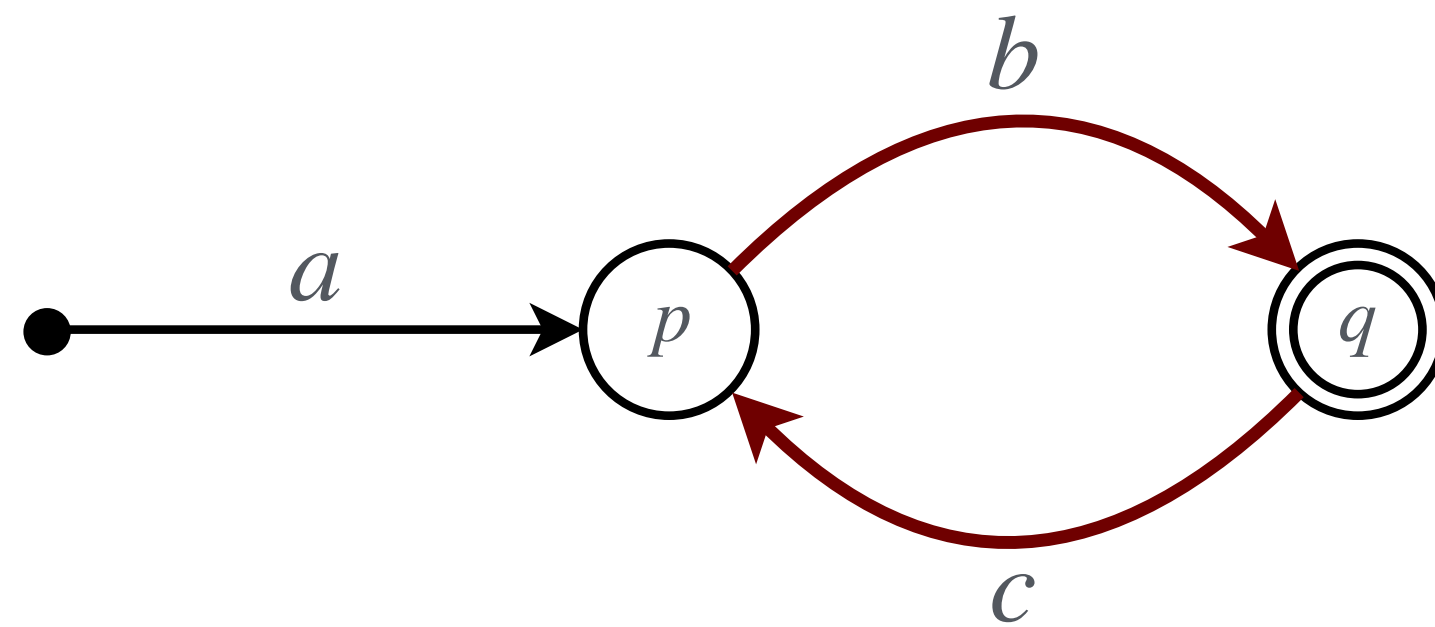
If an action achieves a fact then it's used

$$\forall p \in G$$

Goal facts need to be achieved

MIP Formulation : Acyclicity

- This is not a complete MIP formulation
- We are missing causal acyclicity



MIP Formulation : Static Formulations

- Time Labeling:
 - Assign a time label to each fact
 - Add constraints ensuring time labels of effects are greater of those of preconditions

$$t_p - t_q + 1 \leq |P| (1 - x_{a,q}) \quad \forall a \in A, p \in \text{pre}(a), q \in \text{add}(a)$$

- Compact formulation, but weak LP relaxation
- The resulting formulation will be called **TL(Π)**

MIP Formulation : Static Formulations

- Vertex Elimination:
 - Consider the causal graph $\mathbf{G}_{\Pi}$ of our delete-relaxed task
 - Construct the vertex elimination graph $\mathbf{G}_{\Pi}^*$ (with any elimination ordering) and let Δ be the set of triples added during the elimination process

$$\begin{aligned}x_{a,q} &\leq e_{p,q} && \forall a \in A, p \in \text{pre}(a), q \in \text{add}(a) \\e_{p,q} + e_{q,p} &\leq 1 && \forall (p,q) \in E_{\Pi}^* \\e_{p,q} + e_{q,r} - 1 &\leq e_{p,r} && \forall (p,q,r) \in \Delta\end{aligned}$$

- Can grow large in practice (still polynomial), but stronger LP relaxation
- The resulting formulation will be called $\mathbf{VE}(\Pi)$

MIP Formulation : Dynamic Formulations

- Subtour Elimination Constraints
 - Similarly as in the TSP
 - Find cycles in the causal relation graph of (infeasible) solutions and add violated constraints as lazy constraints

$$\sum_{(p,q) \in C} x_{a,p} \leq |C| - 1$$

- Cycles can be separated in linear time with a graph visit
- The resulting formulation will be called **SEC(Π)**

MIP Formulation : Dynamic Formulations

- Landmark Constraints
 - A disjunctive landmark $\mathbf{L} \subseteq \mathbf{A}$ is a set of actions such that every plan has at least one action from $\mathbf{L}$
 - Delete-Free planning is equivalent to solving hitting set problems over all its landmarks
 - As for SEC, given an infeasible solution we add the violated landmark as lazy constraint

$$\sum_{a \in L} x_a \geq 1$$

- Landmarks can be separated in polynomial time through a combinatorial algorithm
- The resulting formulation will be called **LMC**(Π)

Warm Start

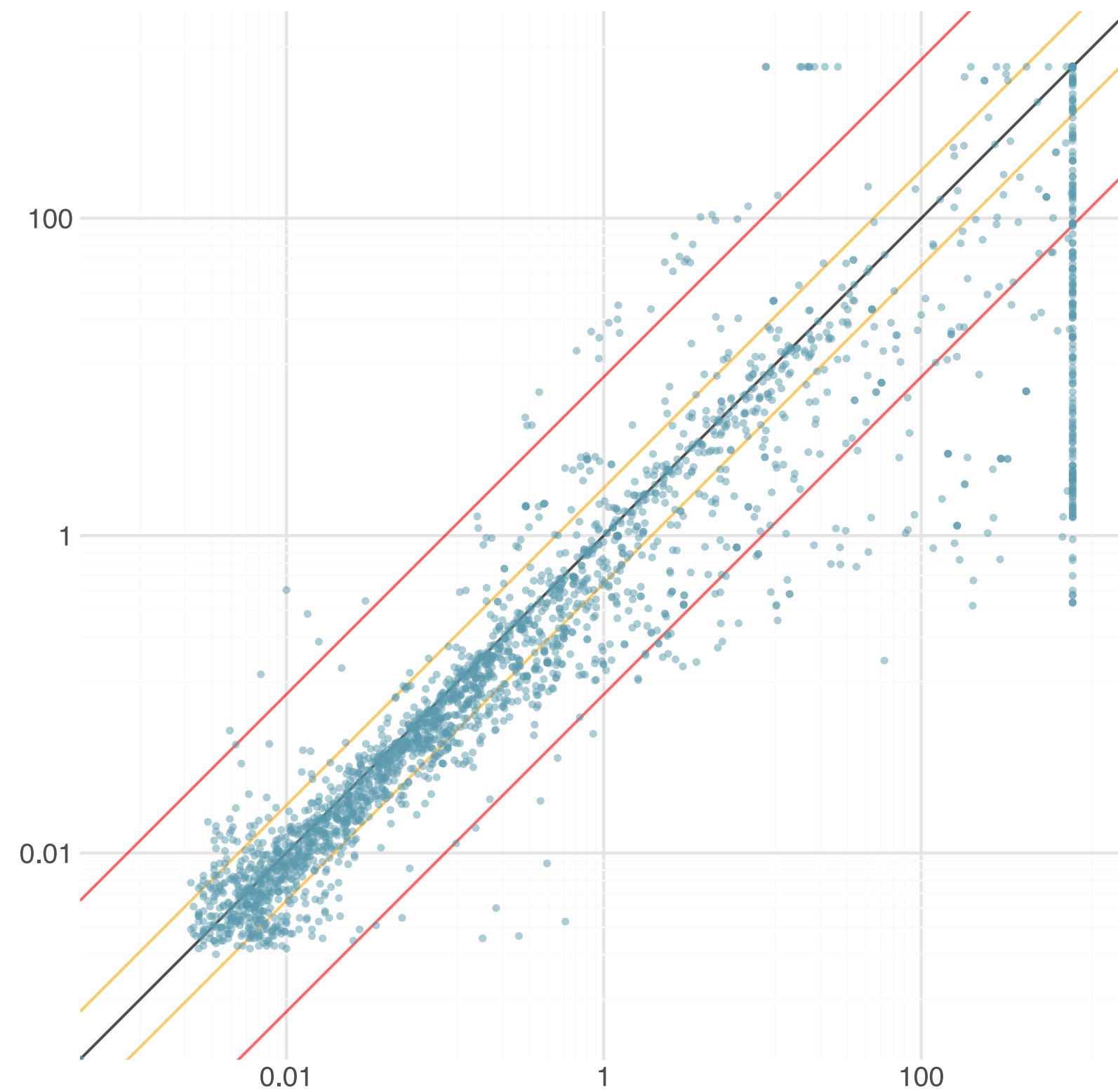
- Primal Heuristic
 - Greedy algorithm with a lookahead strategy based on the h^{add} heuristic to evaluate different actions
 - Runs that uses the primal heuristic will be called with h (es. $\mathbf{LMC}_h(\Pi)$)
- LM-cut initialization
 - LM-cut is a state of the art admissible heuristic for AI Planning
 - Based on the iterative computation of disjunctive landmarks
 - We compute LM-cut and use the landmarks found as an initial set of constraints
 - Runs that uses the LM-cut initialization will be called with $\mathbf{LM}$ (es. $\mathbf{LMC}_{\mathbf{LM}}(\Pi)$)

Fractional Solutions: Landmarks

- We proved that finding whether a violated landmark exists given a fractional solution is NP-hard (reduction from Vertex Cover)
- Formulating an exact method of finding violated landmarks is not practical
- We devised an heuristic approach based on the max-flow problem that if at least one violated landmark exists, in ~97% of the cases we are able to find one
- We are currently working on improving the violated landmark by looking for a minimal one:
 - Using a local search strategy to find better landmarks (given a violated landmark l , choose an action $a \in l$ to round its fractional value to 1, then repeat the separation process)
 - Choose actions by fractionality (closer to 1 first) and failure rate (least failures first)

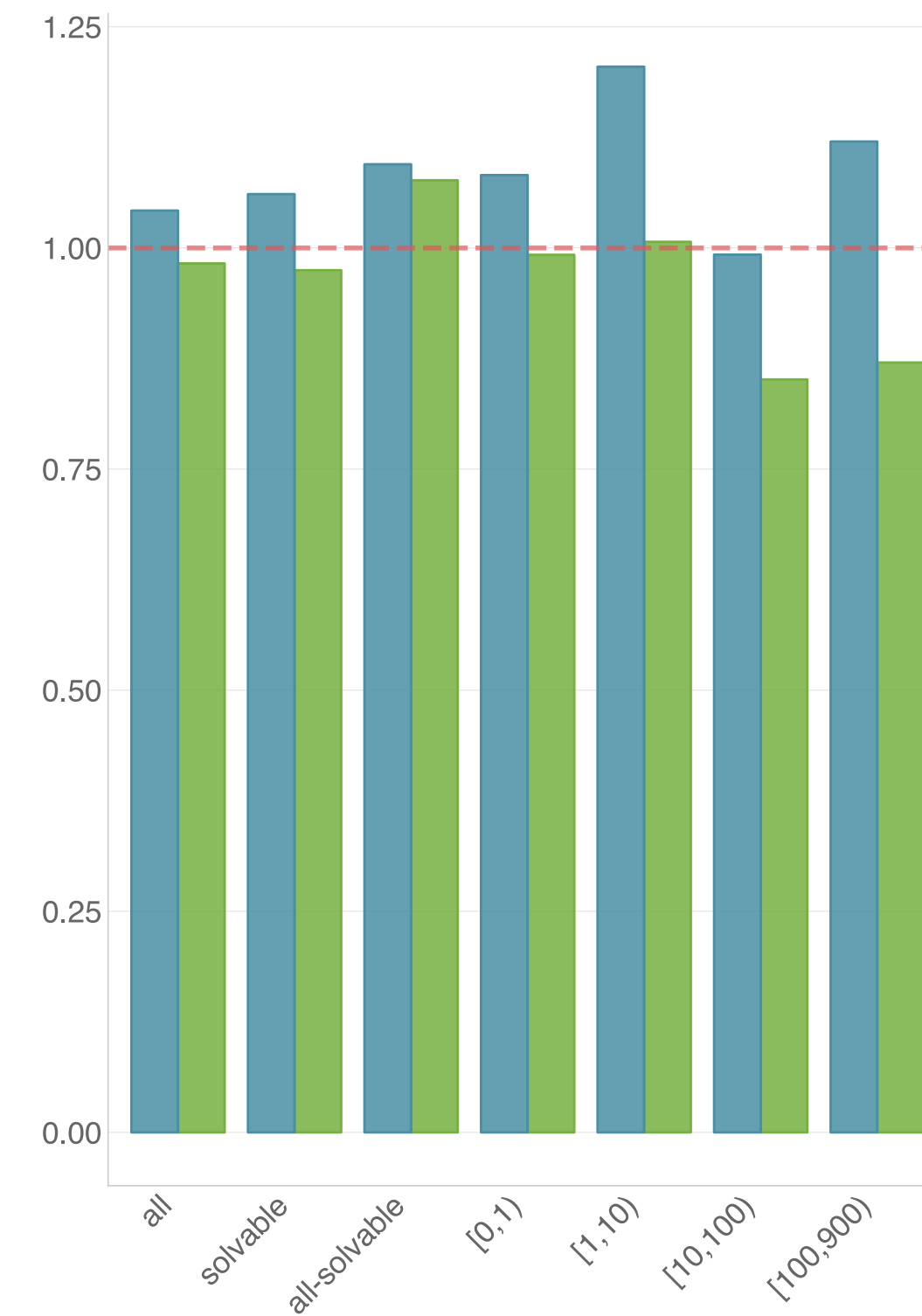
Results

Scatter Plot



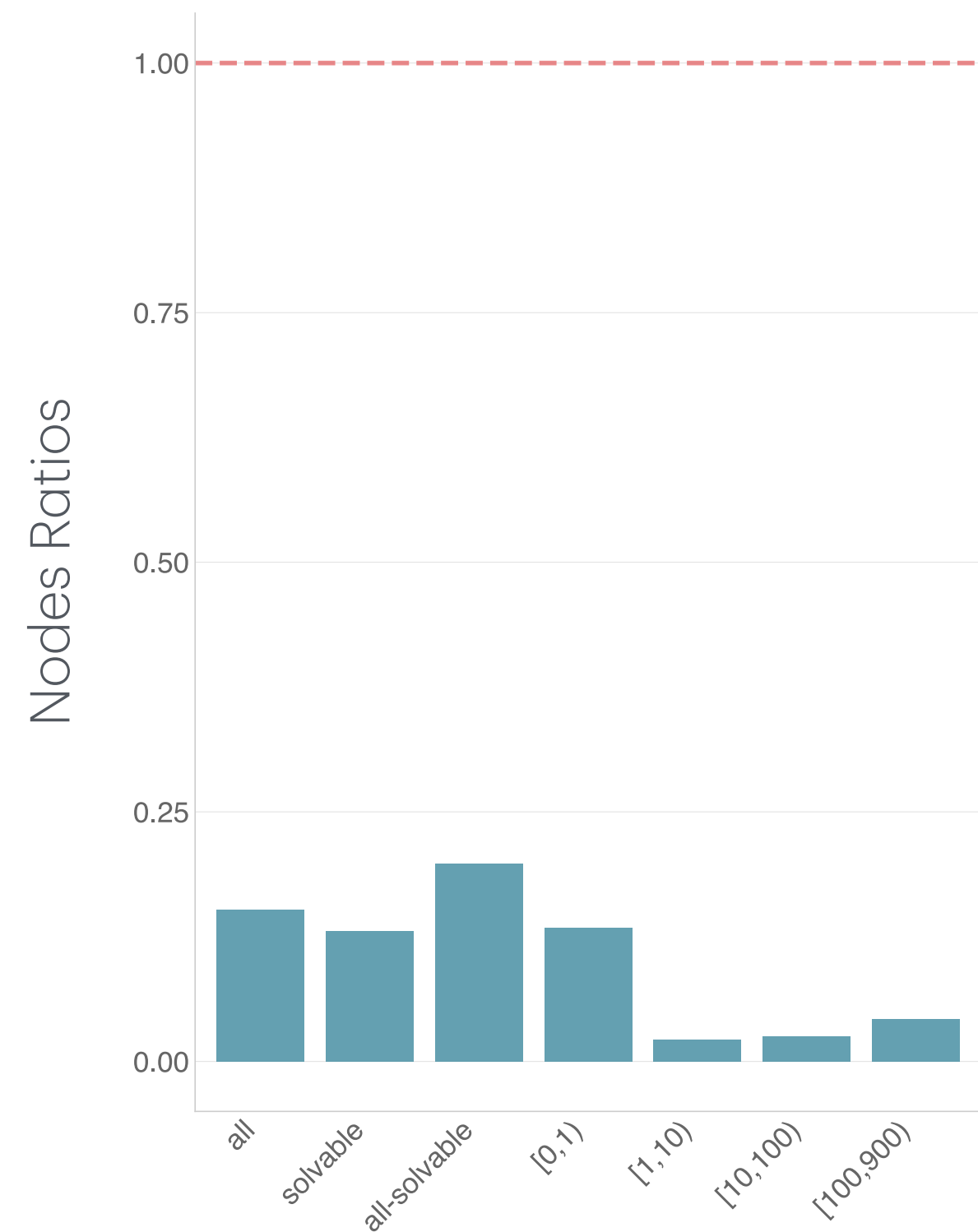
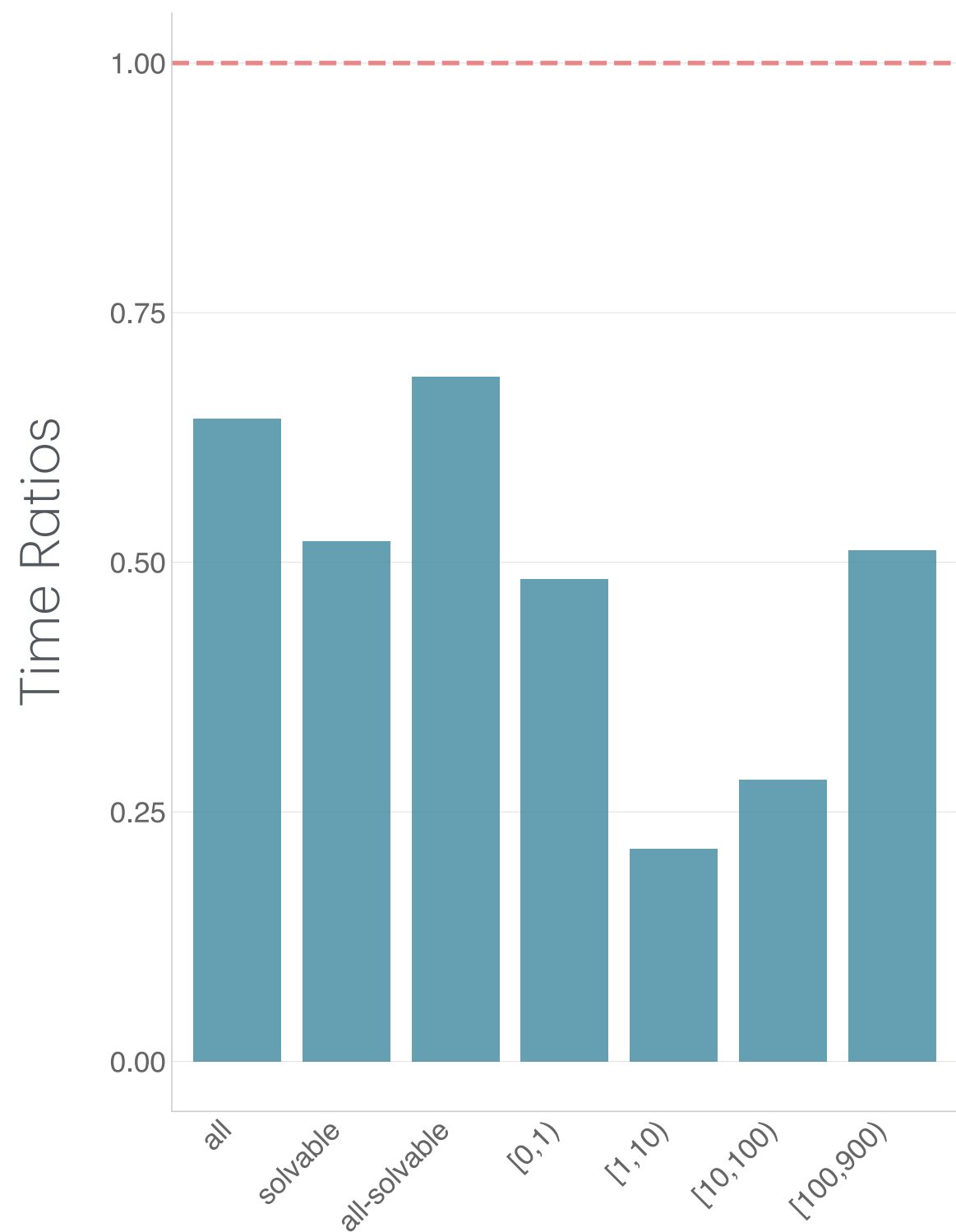
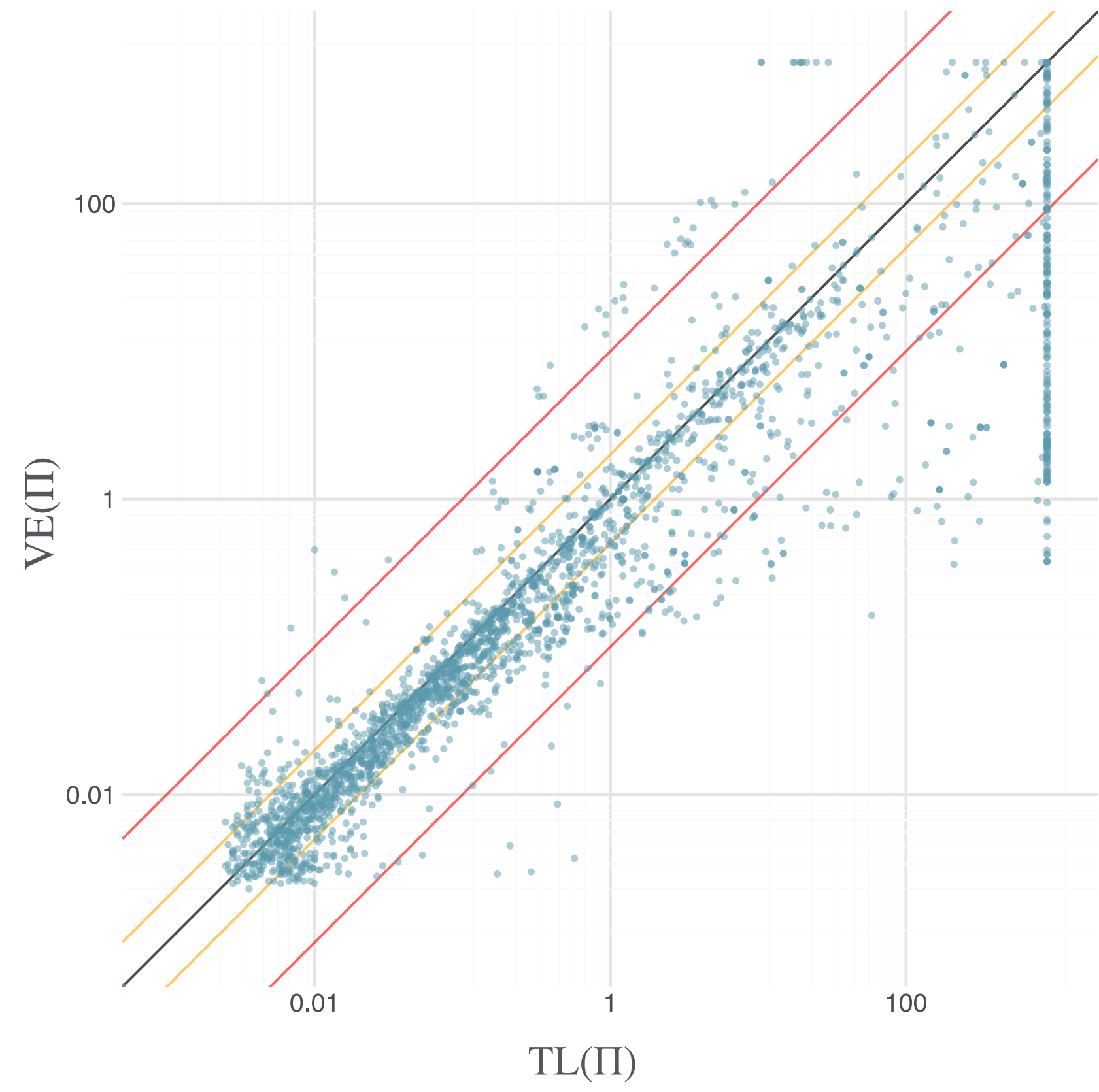
Pairwise time comparison between instances
with speedup reference lines (2x and 10x)

Bar Plot

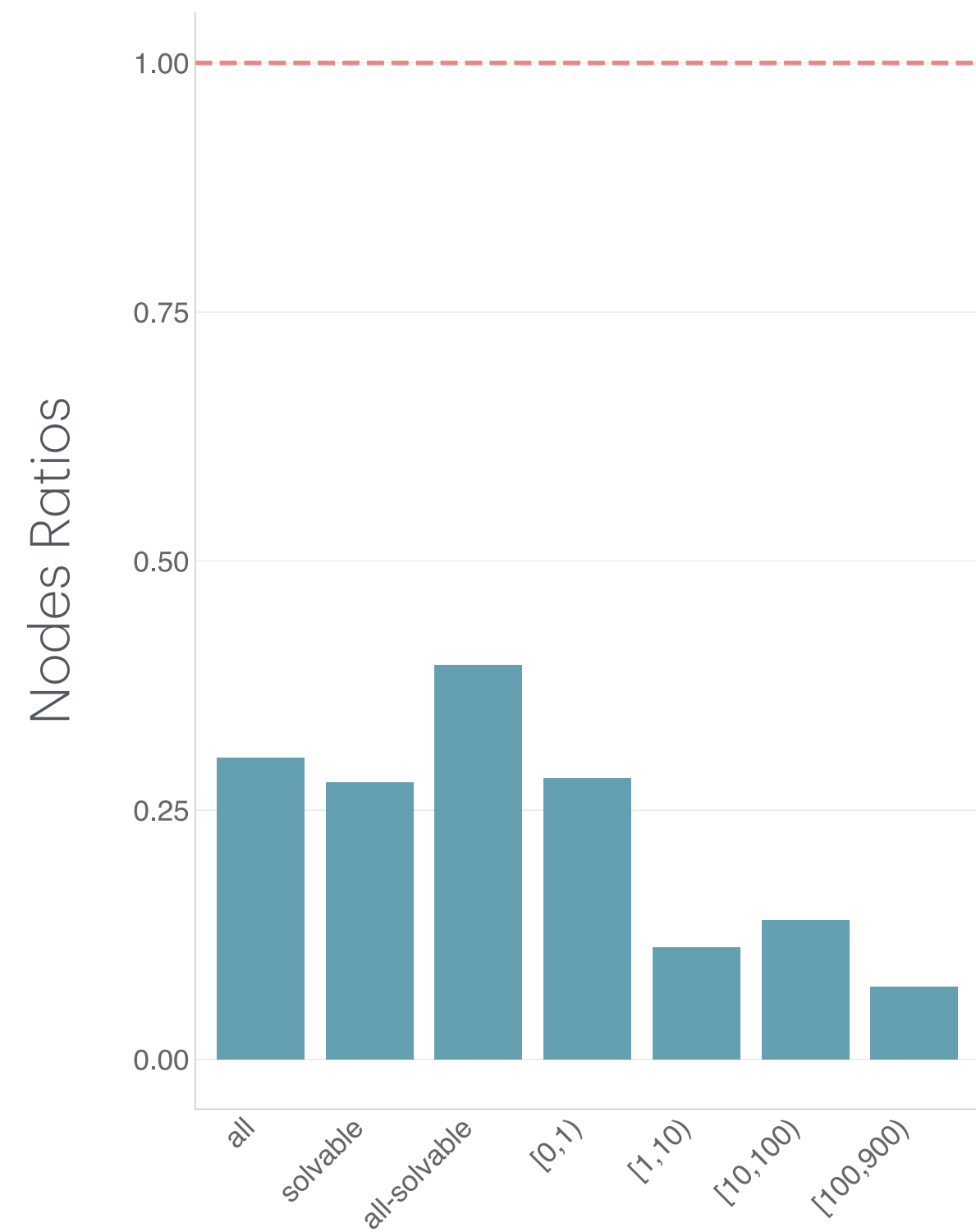
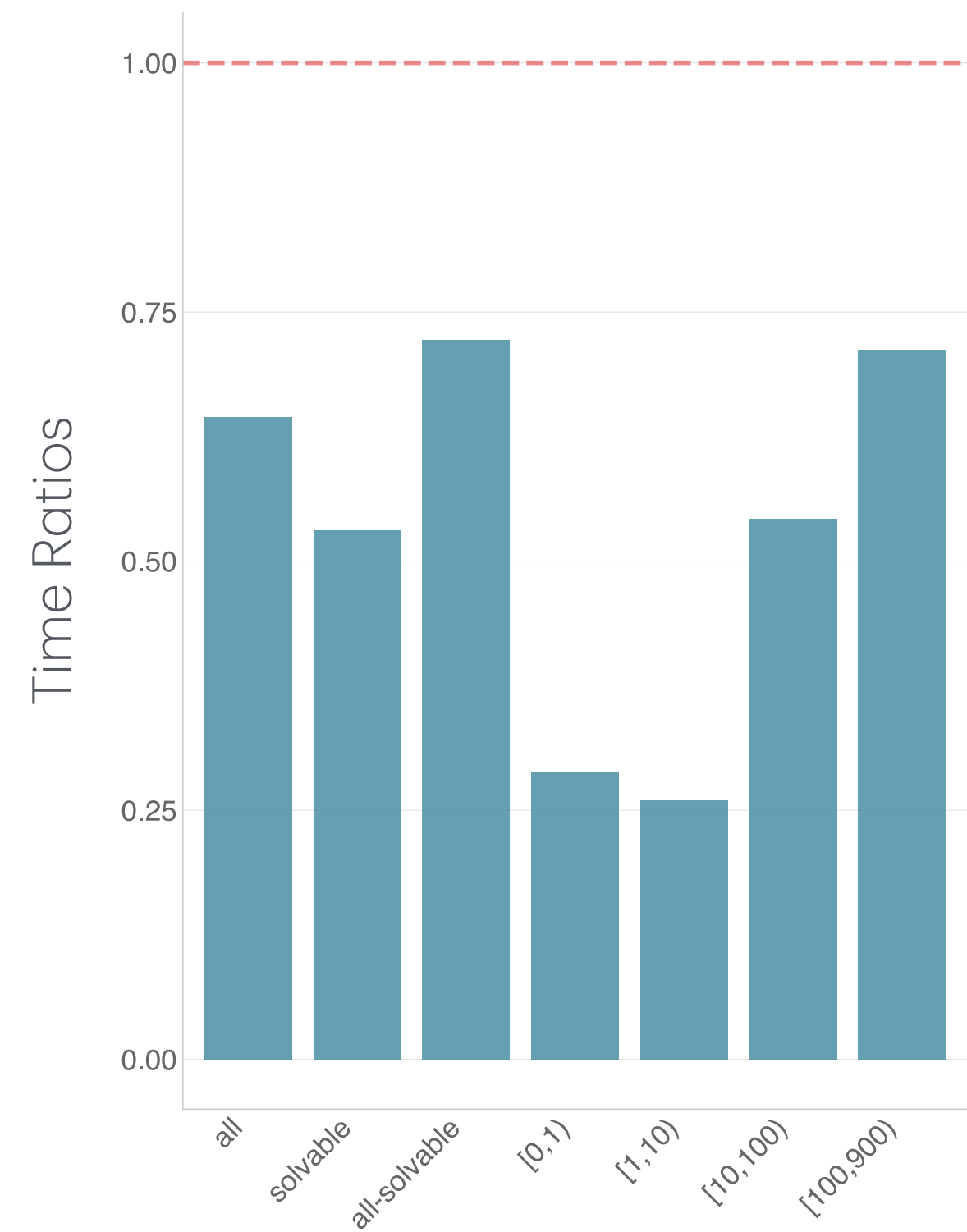
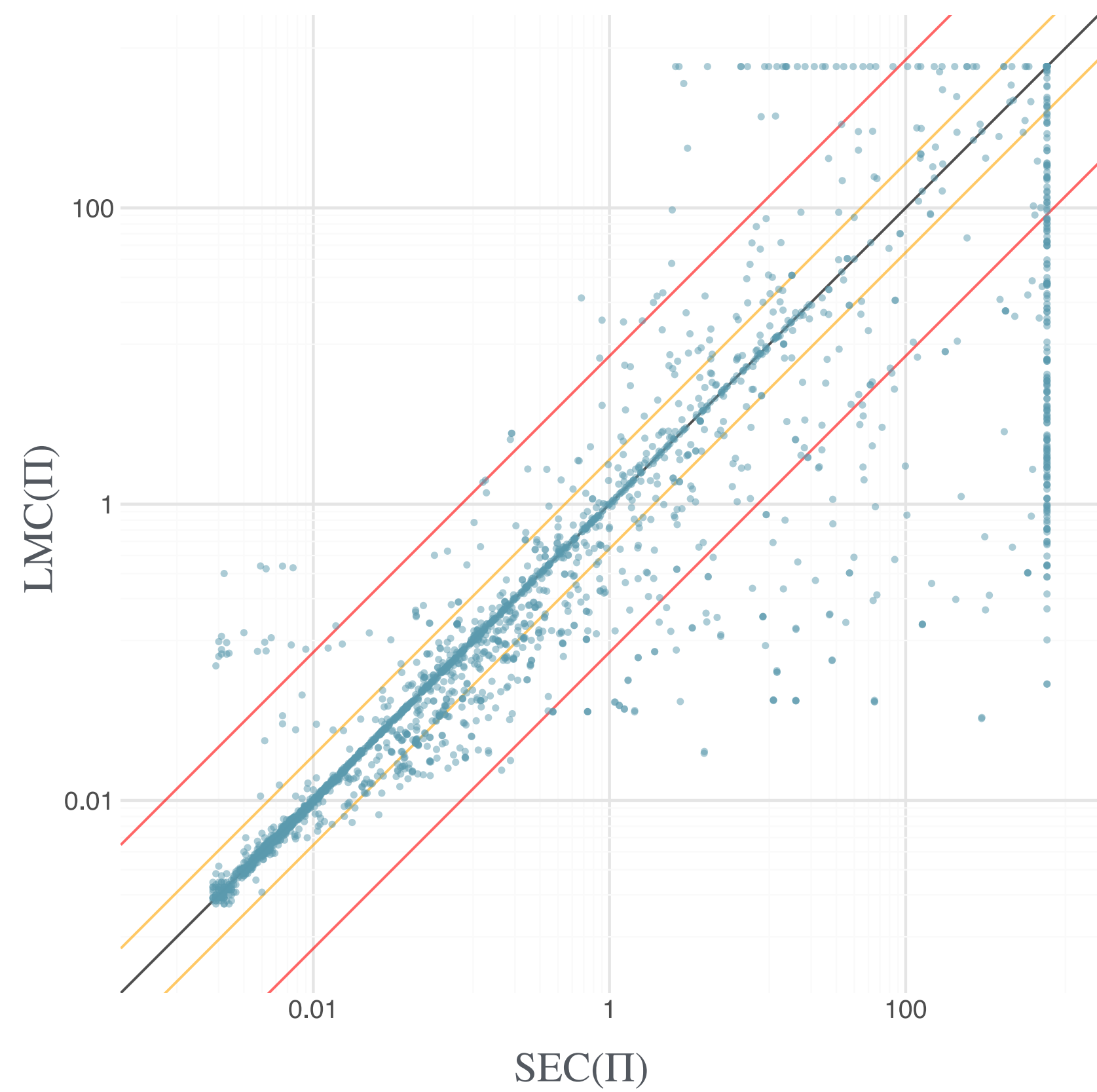


Ratios between baseline model (dashed line)
and proposed models (vertical bars) on a
specific metric (time/nodes)

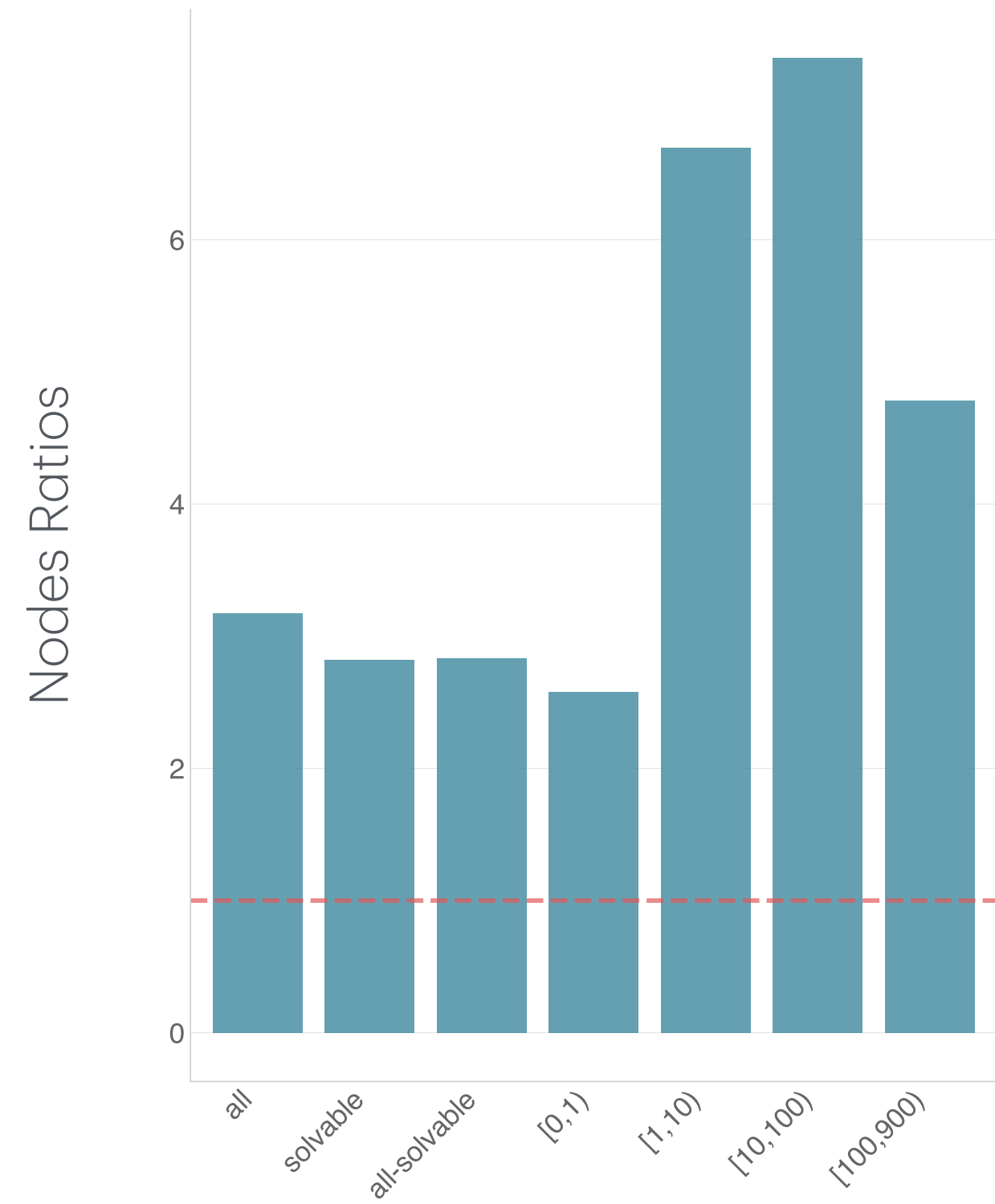
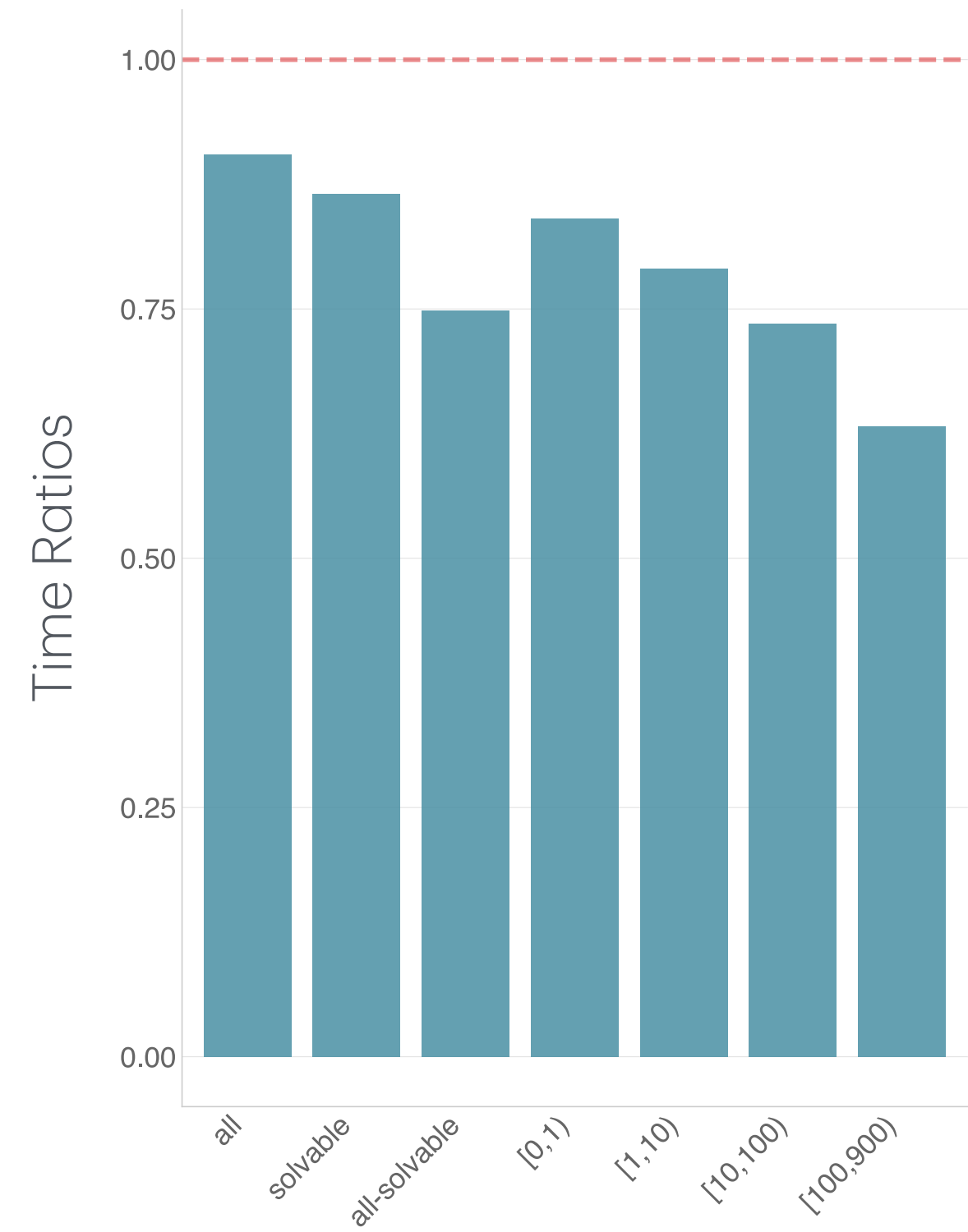
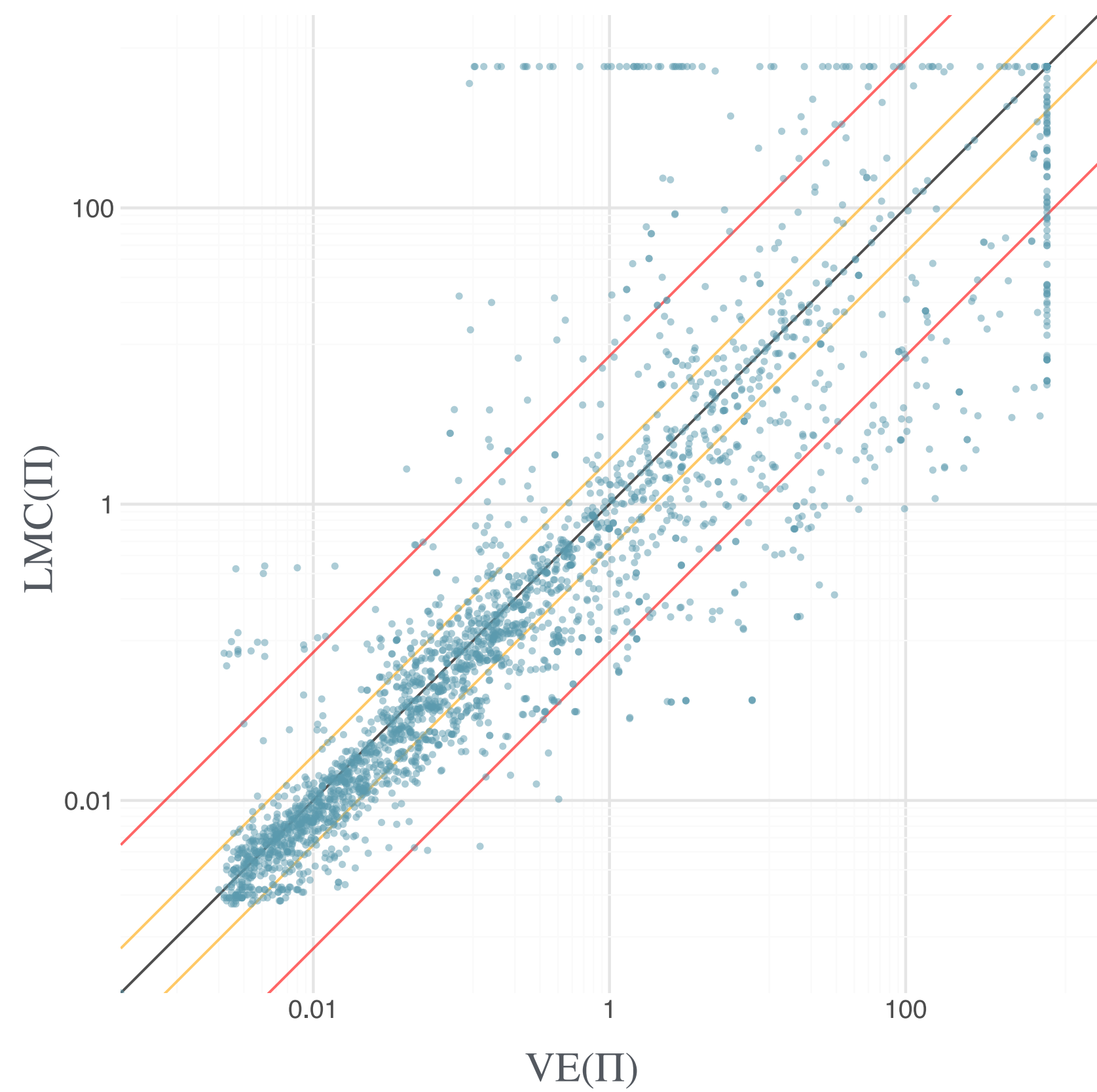
Results: $TL(\Pi)$ vs $VE(\Pi)$



Results: $\text{SEC}(\Pi)$ vs $\text{LMC}(\Pi)$

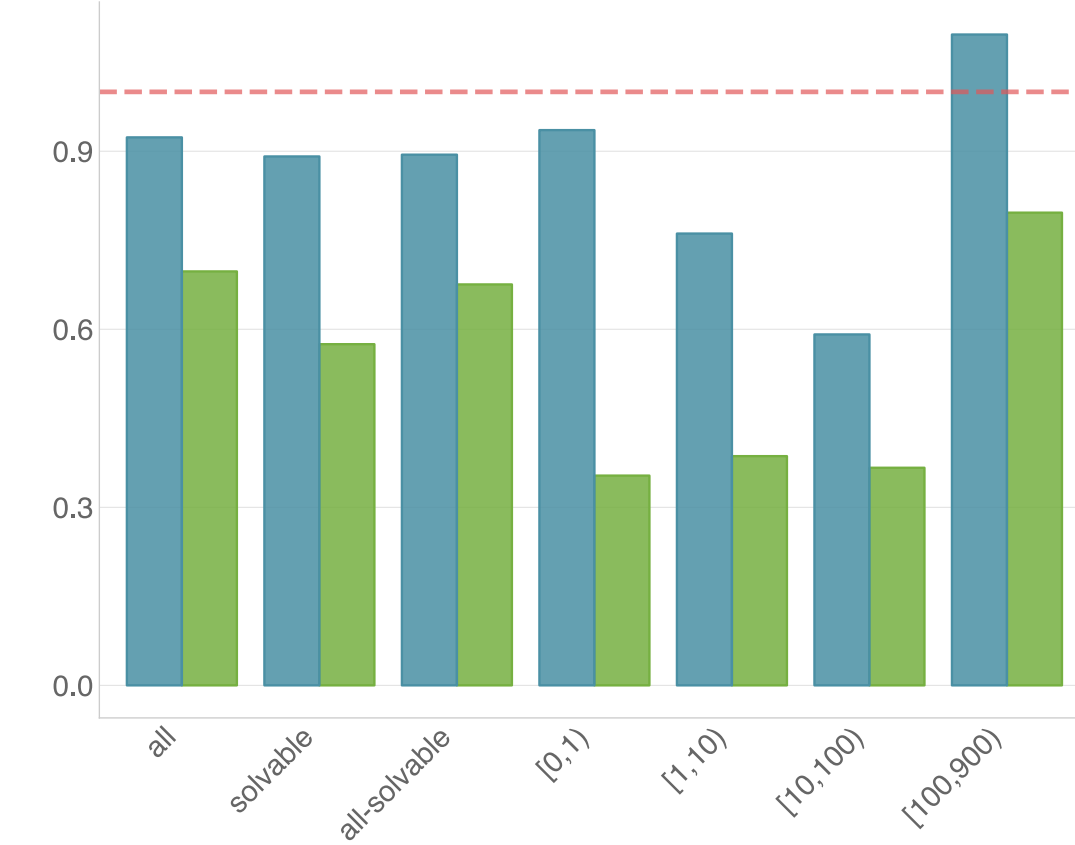
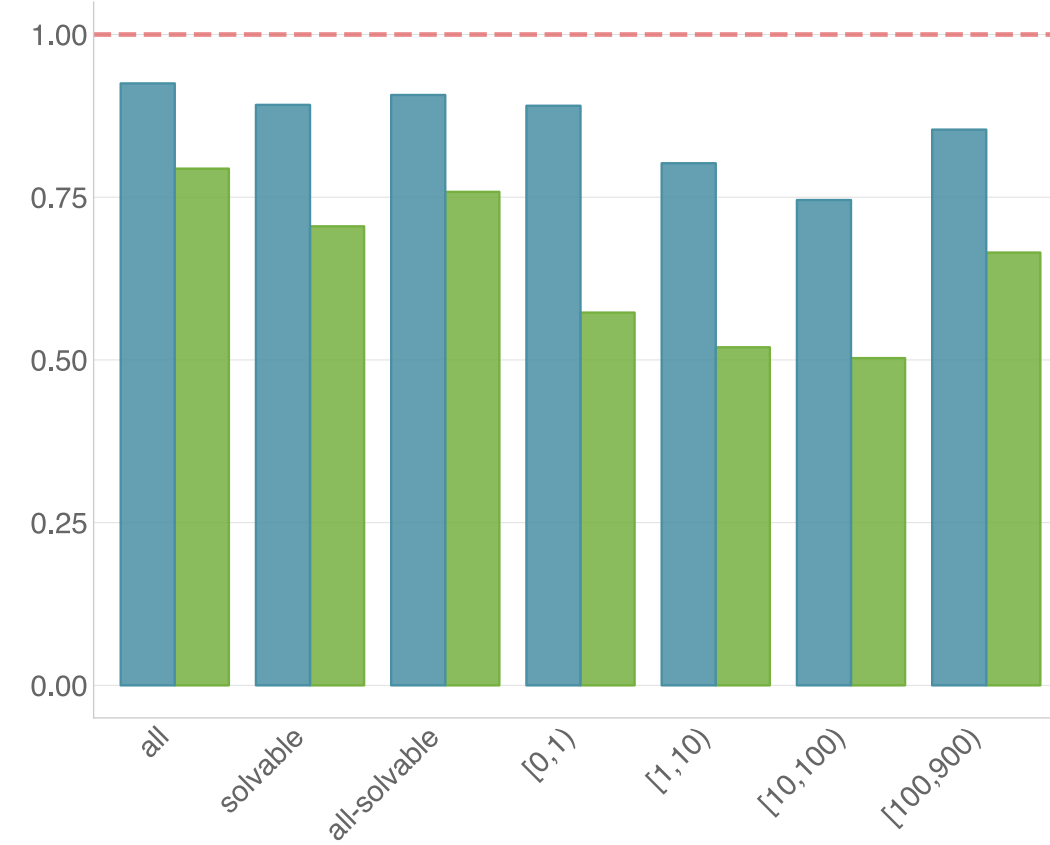
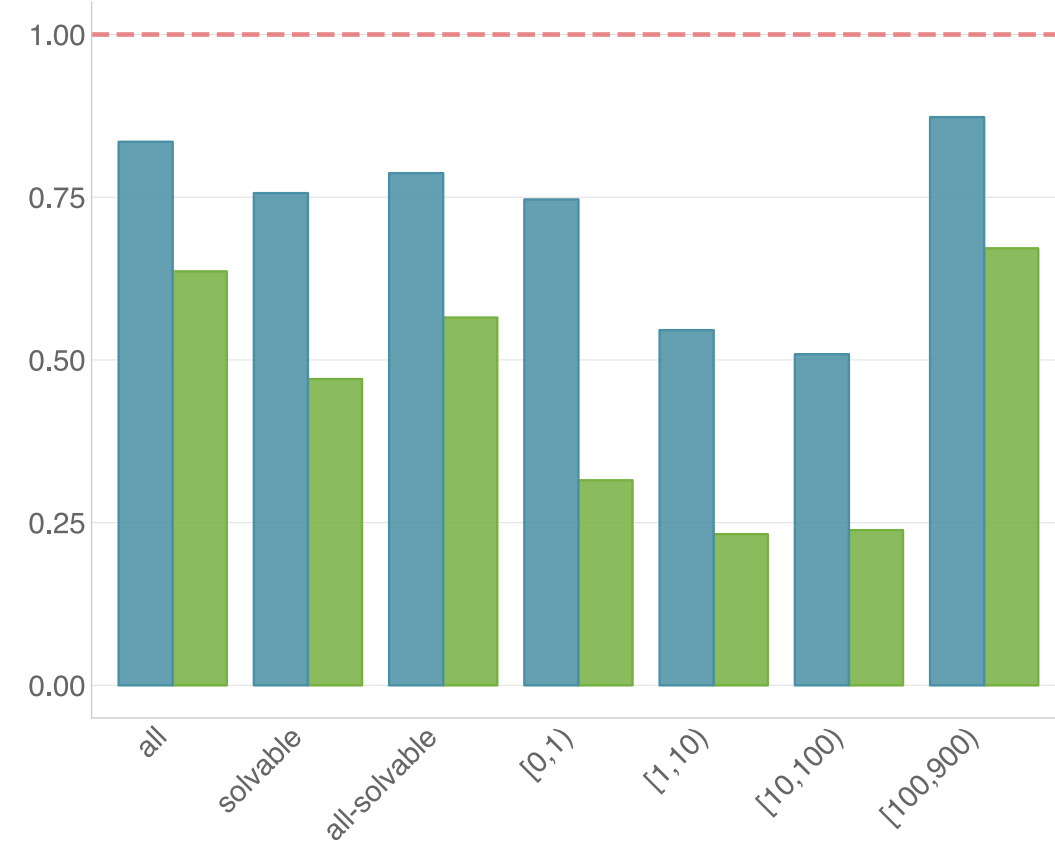


Results: $VE(\Pi)$ vs $LMC(\Pi)$

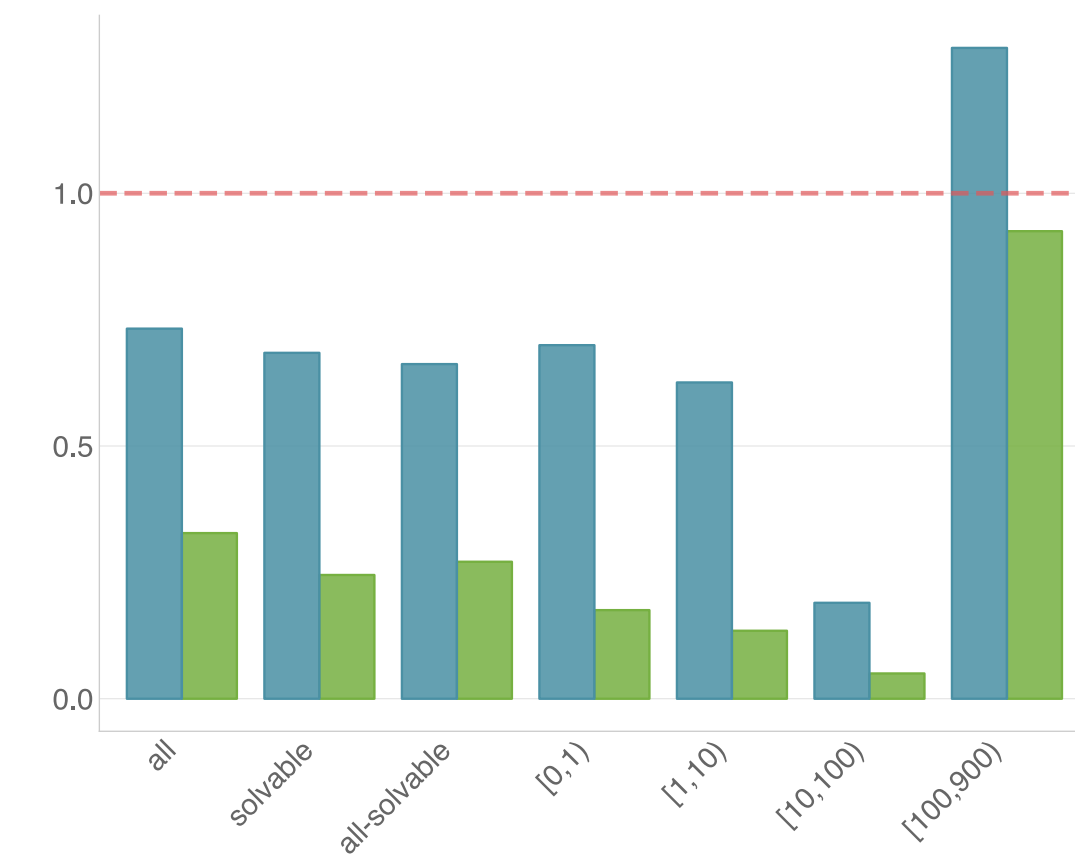
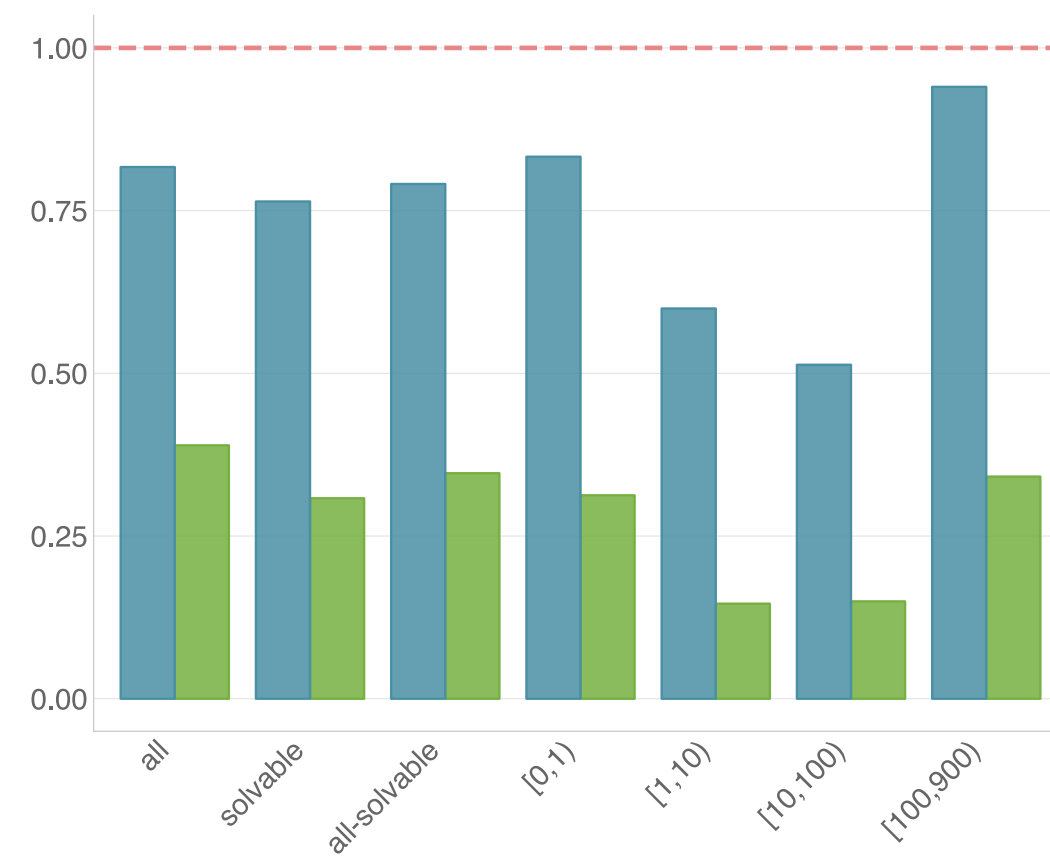
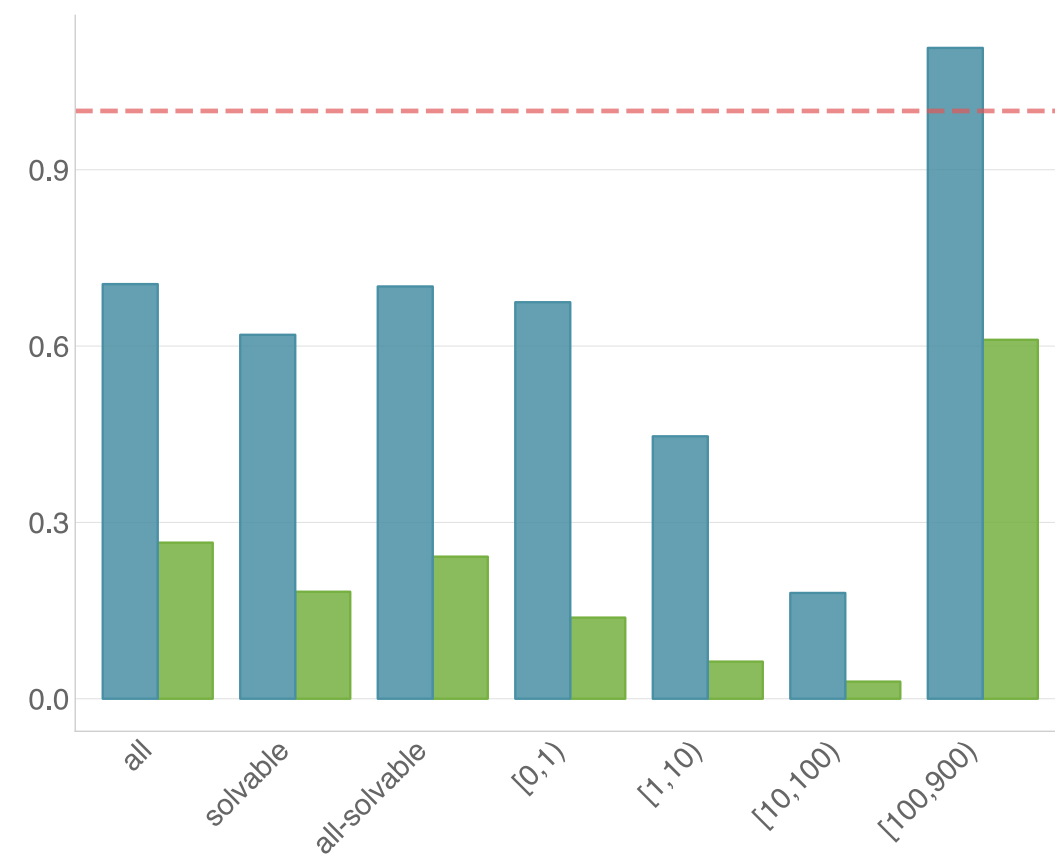


Results: Warm start effects

Time Ratios



Nodes Ratios

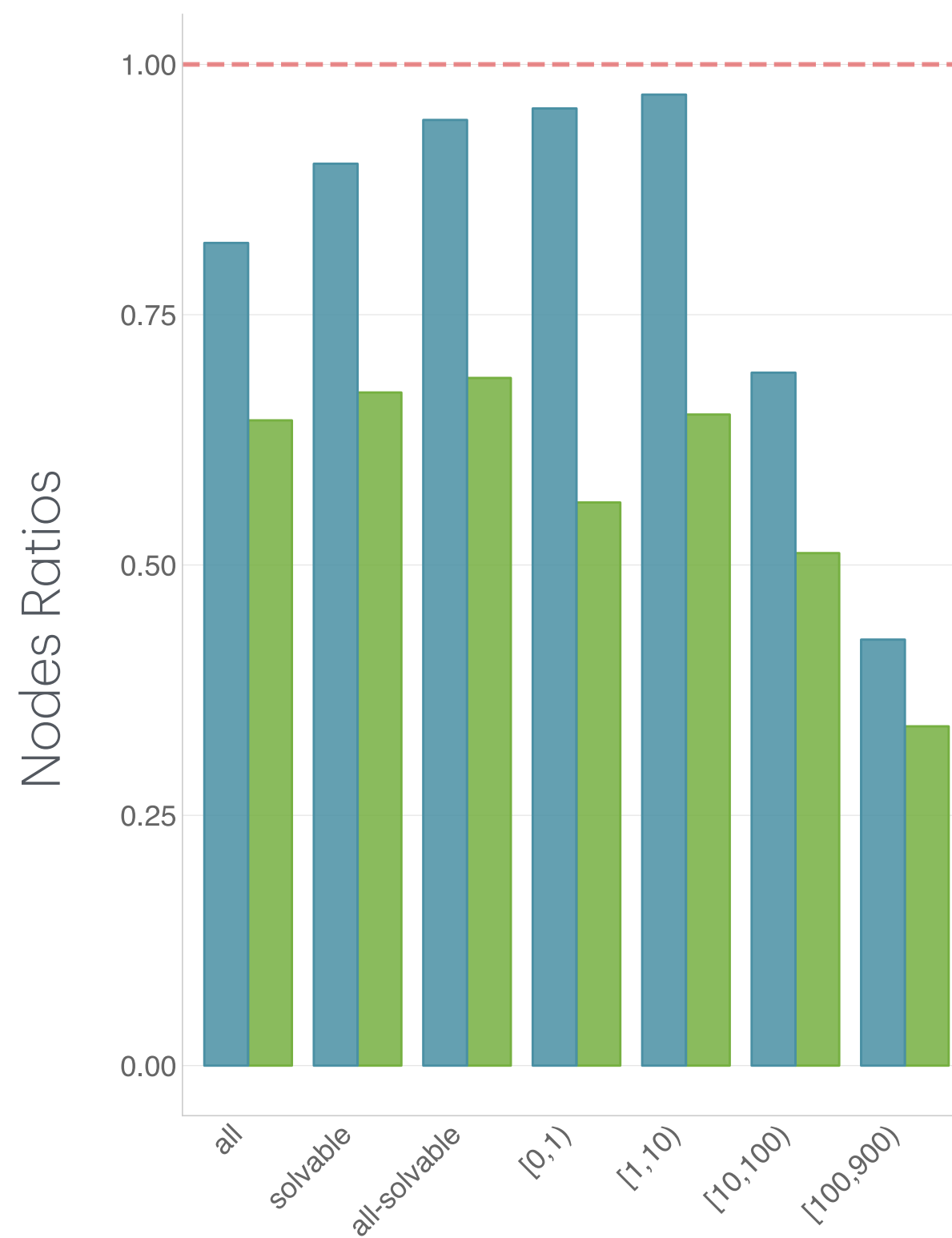
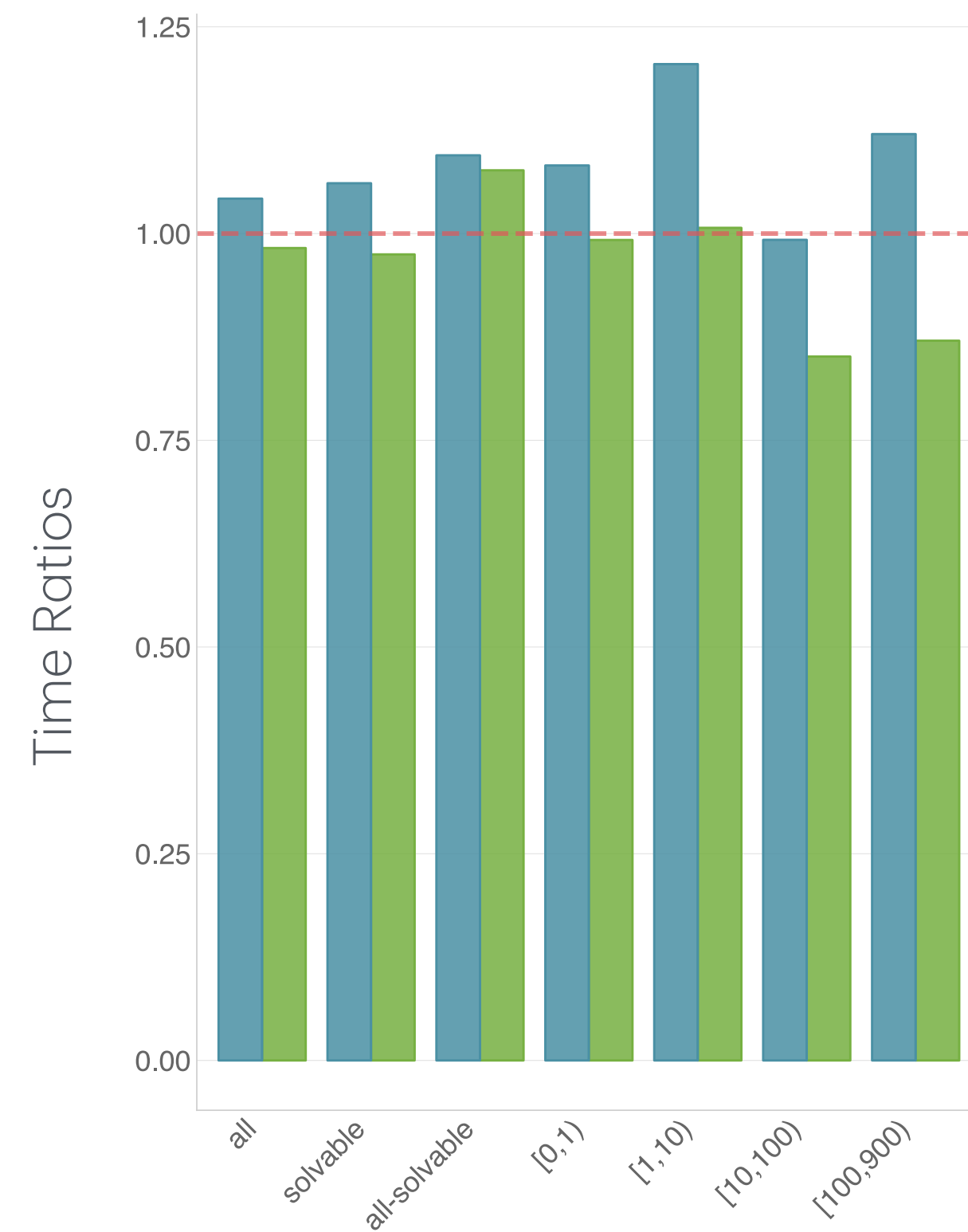


TL(Π) vs TL_h(Π) vs TL_{h,LM}(Π)

VE(Π) vs VE_h(Π) vs VE_{h,LM}(Π)

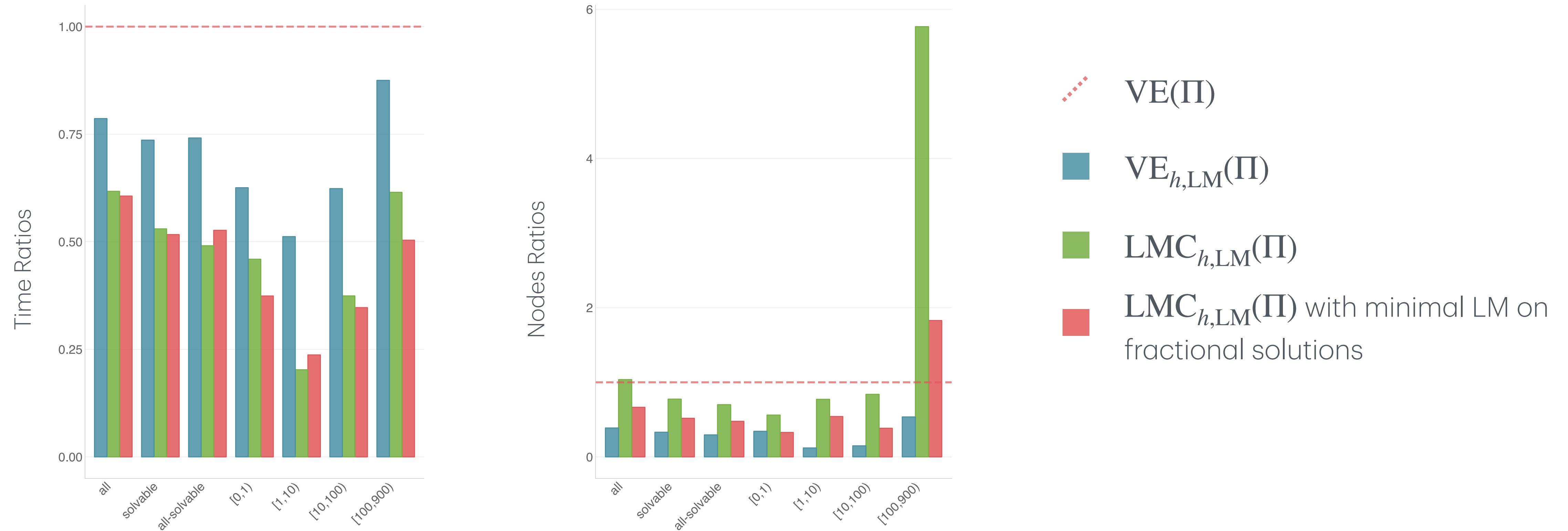
LMC(Π) vs LMC_h(Π) vs LMC_{h,LM}(Π)

Results: Fractional Solutions



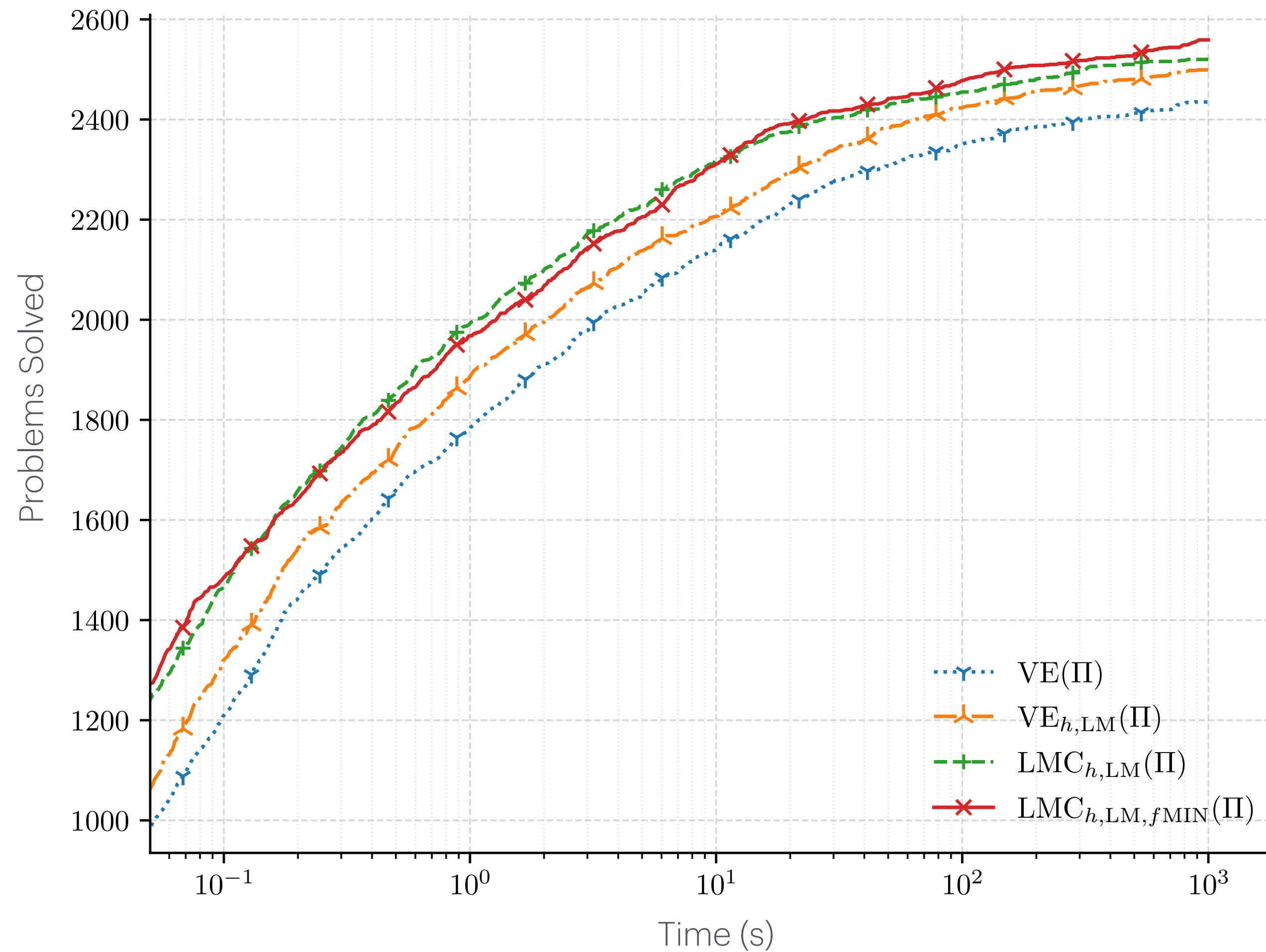
- $\text{LMC}_{h,\text{LM}}(\Pi)$
- $\text{LMC}_{h,\text{LM}}(\Pi)$ with “naive” LM on fractional solutions
- $\text{LMC}_{h,\text{LM}}(\Pi)$ with minimal LM on fractional solutions

Results: Overall Improvements



time limit: 15m - memory limit (CPLEX): 4GB

Results: Overall Improvements



time limit: 15m - memory limit (CPLEX): 4GB

Conclusions

- With the proposed warm start approaches we managed to improve existing static formulations
- Approaches based on SEC didn't prove effective in practice
- Separating violated landmark constraints beats the vertex elimination method and synergizes better with the LM-cut warm start
- The time obtained by separating landmarks from fractional solutions can still be improved, as it's suggested by the significantly lower number of nodes explored
- Maybe there are opportunities on custom branching rules?

Thanks for your attention!

If you have any questions, feel free to contact me: matteo.zanella.3@phd.unipd.it